

The Owner's Dilemma: Testing Transfer-Window Signals for Sporting and Financial Decisions

Soccer

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Abstract.

Football owners face a recurring trade-off between monetizing player value and preserving the squad strength needed for immediate competition. Sporting competitiveness and financial return therefore belong in the same decision. In principle, efficient choices lie on a Pareto frontier, where improving one objective further requires giving up some of the other. Which point an owner prefers depends on the club's priorities and constraints. Estimating that frontier credibly, however, requires both a reliable sporting-response model and realistic alternative transfer-window scenarios. We test whether aggregate summer-window financial and activity signals improve forecasts of associated-season La Liga points per game (PPG) beyond transparent measures of prior strength.

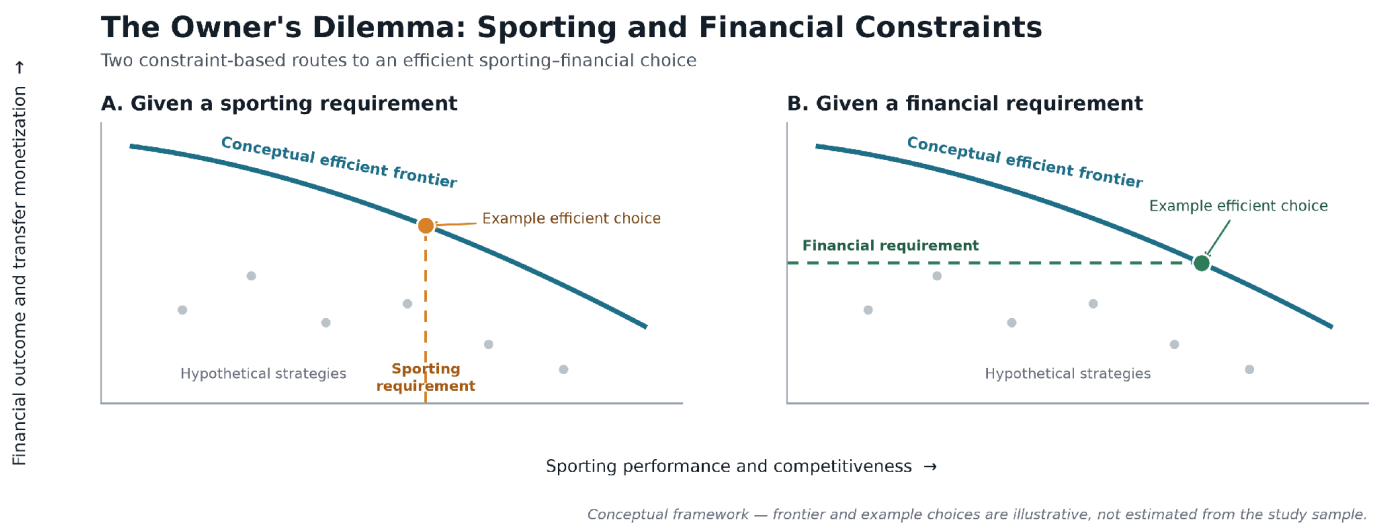


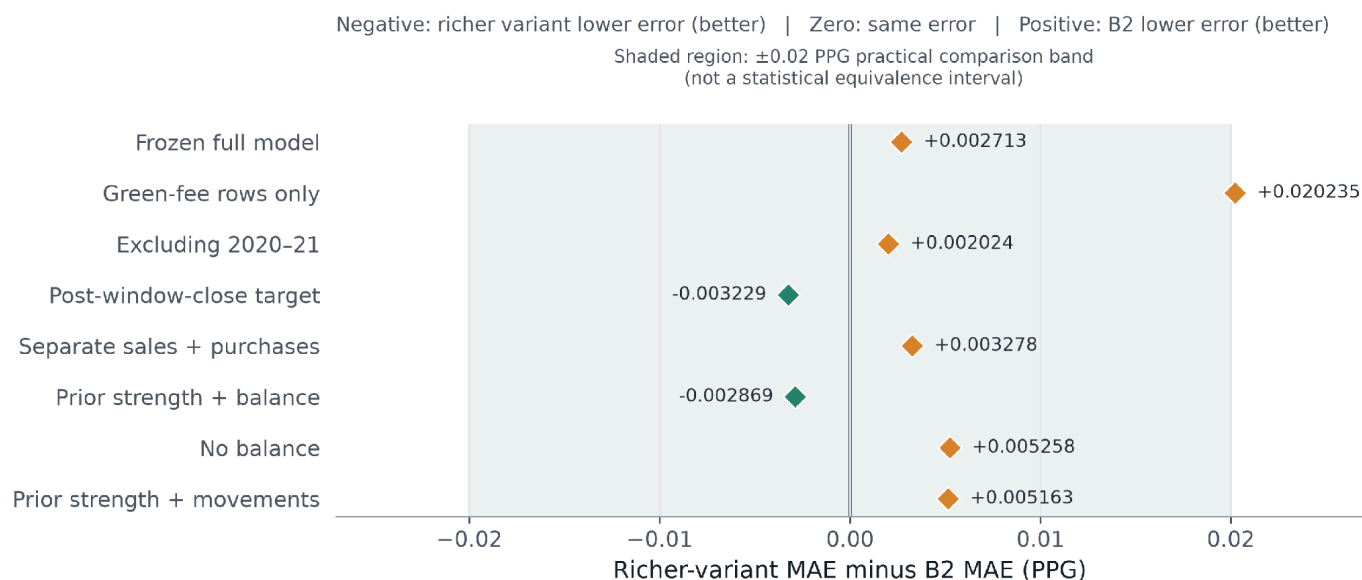
Figure 1. Conceptual representation of the Owner's Dilemma. Left: given a sporting requirement, the frontier identifies the highest financial outcome among efficient combinations. Right: given a financial requirement, it identifies the strongest sporting outcome among efficient combinations. The frontier, hypothetical strategies, and highlighted choices are illustrative and were not estimated from the study sample.

Our analysis covers 159 La Liga club-by-summer-window observations across eight seasons, from 2017-18 through 2024-25. The analysis combines sporting outcomes with a published transfer-transaction dataset. The chronological evaluation design uses 99 training observations, 40 validation observations, and a 20-club held-out test from 2024-25. Test outcomes were not used while features, preprocessing choices, evaluation metrics, baselines, and the model specification were being finalized.

Our financial measure is known-fee transfer balance - known sales fees minus known purchase fees - not accounting profit. It excludes wages, amortization, agent fees, bonuses, payment timing, and other costs. We compare mean PPG, persistence, and a simple prior-strength baseline (B2) with a richer ordinary-least-squares model using prior strength, promotion status, known-fee balance, movement counts, fee coverage, and COVID-era context. Mean absolute error (MAE) in PPG is the primary metric. Because league play can begin before the summer window closes, a post-window-close target is treated only as a post-hoc sensitivity.

Validation favors B2 (MAE 0.219931 versus 0.226723). On the held-out test, mean PPG records 0.361310 MAE, persistence 0.271571, B2 0.238262, and the richer model 0.240975. The richer model improves on persistence by 0.030596 PPG MAE but is 0.002713 worse than B2. Post-hoc checks tell the same broad story: no core variant shows a stable material advantage over B2 within the ± 0.02 PPG practical comparison band. That band is descriptive, not a statistical equivalence interval, and these checks are not independent validation.

Post-hoc robustness relative to B2



Post-hoc robustness checks — not independent validation.

Figure 2. Post-hoc robustness of richer transfer-window variants relative to B2. Values are variant MAE minus B2 MAE: negative favors the variant; positive favors B2. The shaded ± 0.02 PPG region is a practical comparison band, not a statistical equivalence interval. These checks are not independent validation, and they show no stable material advantage over B2.

Aggregate transfer-window totals therefore provide useful context and improve on naive persistence, but they do not reliably beat a compact prior-strength baseline. On this evidence, they cannot support an owner-facing sell/retain optimizer or an empirical efficient frontier. The study is predictive and observational: it identifies no causal transfer effect, optimal sale amount, safe-to-sell threshold, or validated alternative-window scenario generator. Evidence is also limited to one 20-club held-out season. A genuine optimization framework would need richer player-level departures and arrivals, quality, roles, minutes, replacement strength, transfer type, contract context, financial detail, and realistic counterfactual windows. The practical result is a clear boundary for the Owner's Dilemma: aggregate window-level activity alone is not enough to make the sporting-financial choice owners ultimately care about.

This research was developed in collaboration with SoccerSolver. SoccerSolver currently works with more than 10 football clubs.

Open-source repository: <https://github.com/RupayanHalder39/MITSloanProblem3-Owners-Dilemma-Football>